

File Handling & Data Processing System

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Internship

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Introduction

Managing large amounts of data manually can be time-consuming and prone to errors. Organizations often rely on automated systems to store, process, analyze, and generate reports from their data efficiently. File handling is one of the fundamental concepts in Python that enables developers to work with files such as CSV, text, and log files.

The **File Handling & Data Processing System** is a Python-based console application developed to demonstrate practical file handling and data processing techniques using CSV files. It allows users to generate sample sales data, load records, search products, filter and sort information, generate sales summaries, and export processed data into CSV and text report files.

Unlike a basic file handling program, this project also includes automatic directory creation, activity logging, data filtering, sorting, searching, report generation, and exception handling. The application stores all data locally and requires only Python's built-in libraries, making it lightweight, portable, and easy to use without any external database or third-party packages.

This project helped me strengthen my understanding of Python file handling, CSV processing, logging, exception handling, modular programming, and basic data analysis while developing a practical real-world application.

Objectives

The main objectives of this project are:

- To develop a complete file handling system using Python.
- To read and process data from CSV files efficiently.
- To perform searching, filtering, and sorting operations.
- To generate meaningful sales summaries and reports.
- To export processed records into CSV and text files.
- To maintain application activity logs using Python logging.
- To implement robust input validation and exception handling.
- To improve programming skills using functions, modules, and file management.

Features

Sales Data Management

- Generate sample sales data automatically.
- Load sales records from CSV files.
- Display records in a structured table.
- Skip invalid or corrupted records safely.

Data Search and Filtering

- Search products by product name.
- Filter records using any available column.
- Support text-based and numeric filtering.
- Display matching records instantly.

Data Sorting

- Sort records using any column.
- Support ascending and descending order.
- Organize records for easier analysis.

Sales Analytics

- Calculate total revenue.
- Calculate total quantity sold.
- Calculate average unit price.
- Identify the highest revenue-generating product.
- Generate category-wise sales summaries.

Export Features

- Export processed records to CSV.
- Generate summary reports in text format.
- Save reports for future reference.

Logging

- Record application activities with timestamps.
- Maintain detailed log files.
- Track important file operations.

Additional Features

- User-friendly menu-driven interface.
- Automatic directory creation.
- Input validation.
- Exception handling.
- Local data storage.

Technologies Used

Technology	Purpose
Python 3	Programming Language
CSV	Data Storage
VS Code	Code Editor
Git	Version Control
GitHub	Project Repository

Python Modules Used

csv

Used for reading, writing, and processing CSV files containing sales records.

os

Used for creating directories, checking file paths, and managing file operations.

logging

Used to record application activities with timestamps for monitoring and debugging.

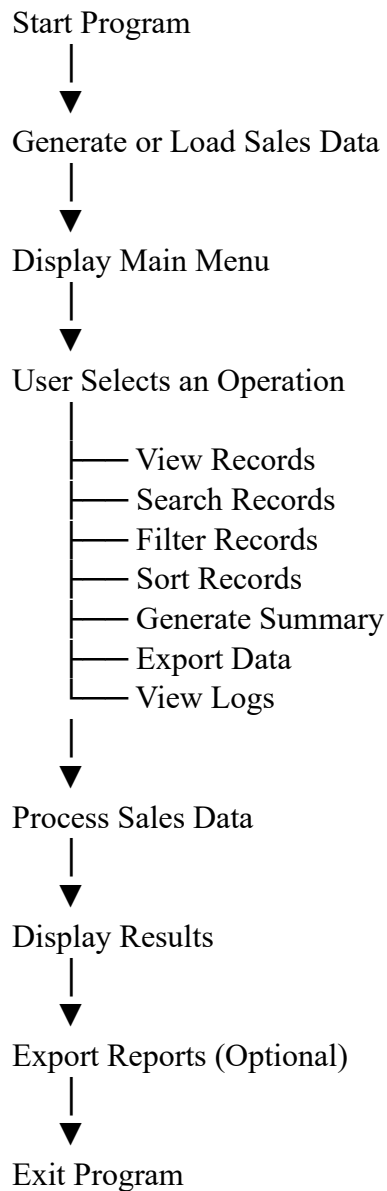
datetime

Used to generate timestamps for reports and log files.

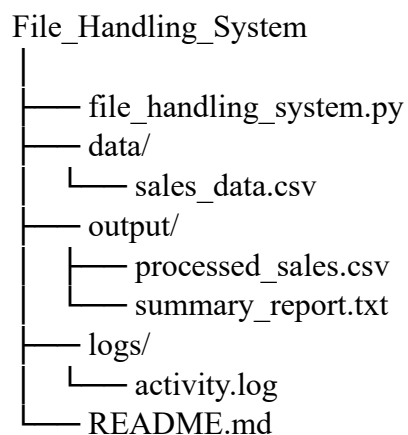
statistics (mean)

Used to calculate the average unit price during sales analysis.

Project Workflow



Project Structure



Screenshots

```
PS C:\Users\rajes\Desktop\python\File Handling> python file_handling_system.py

=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====

Enter your choice: 1
Sample data created at: C:\Users\rajes\Desktop\python\File Handling\data\sales_data.csv

=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====

Enter your choice: 2
Press Enter to use default path (sales_data.csv) or type a full path:
Loaded 10 records from 'sales_data.csv'.

=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====

Enter your choice: 3
ID      Product                Category  Region  Qty  Price  Date
-----
P001    Wireless Mouse           Electronics South   25   599.00 2026-01-05
P002    Office Chair              Furniture  North   10  3499.00 2026-01-06
P003    Bluetooth Speaker        Electronics East    40  1299.00 2026-01-07
P004    Notebook Pack             Stationery West    100   89.00 2026-01-08
P005    LED Desk Lamp             Furniture  South   15   799.00 2026-01-09
P006    USB-C Cable               Electronics North   60   199.00 2026-01-10
P007    Whiteboard Marker Set     Stationery East    75   149.00 2026-01-11
P008    Study Table               Furniture  West    5   4999.00 2026-01-12
P009    Mechanical Keyboard       Electronics South   20  2199.00 2026-01-13
P010    Sticky Notes              Stationery North  150   45.00 2026-01-14

(10 of 10 record(s) shown)
```

```
=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====

Enter your choice: 4
Available columns: ProductID, ProductName, Category, Region, Quantity, UnitPrice, Date
Filter by column: ProductID
Value to match: P001
Found 1 matching record(s).
ID      Product      Category  Region Qty  Price  Date
-----
P001    Wireless Mouse    Electronics South   25   599.00 2026-01-05

(1 of 1 record(s) shown)

Set this as current working set? (y/n): Y
```

```
=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====

Enter your choice: 5
Available columns: ProductID, ProductName, Category, Region, Quantity, UnitPrice, Date
Sort by column: ProductName
Descending? (y/n): y
ID      Product      Category  Region Qty  Price  Date
-----
P001    Wireless Mouse    Electronics South   25   599.00 2026-01-05

(1 of 1 record(s) shown)
```

```
=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====

Enter your choice: 6

----- SALES SUMMARY -----
Total Records      : 1
Total Quantity Sold: 25
Total Revenue      : Rs. 14975.00
Average Unit Price: Rs. 599.00
Top Product        : Wireless Mouse (Rs. 14975.00)

By Category:
  Electronics Qty: 25   Revenue: Rs. 14975.00
-----
```

```
=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====
```

```
Enter your choice: 7
Enter product name keyword: Wireless Mouse
Found 1 record(s) matching 'Wireless Mouse'.
ID   Product      Category   Region  Qty   Price   Date
-----
P001 Wireless Mouse Electronics South   25   599.00  2026-01-05

(1 of 1 record(s) shown)
```

```
=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====
```

```
Enter your choice: 8
Press Enter for default filename (processed_sales.csv) or type a new one:
Exported 1 record(s) to: C:\Users\rajes\Desktop\python\File Handling\output\processed_sales.csv
```

```
=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====
```

```
Enter your choice: 9
Summary report saved to: C:\Users\rajes\Desktop\python\File Handling\output\summary_report.txt
```

```
=====
FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=====
1. Generate sample data file
2. Load data from CSV
3. View all records
4. Filter records
5. Sort records
6. Summarize data
7. Search records by product name
8. Export current records to CSV
9. Export summary report to TXT
0. Exit
=====
```

```
Enter your choice: 0
Goodbye!
```


Sample Output

FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA

1. Generate Sample Data
2. Load Data from CSV
3. View All Records
4. Filter Records
5. Sort Records
6. Summarize Data
7. Search Records by Product Name
8. Export Current Records to CSV
9. Export Summary Report to TXT
0. Exit

Enter your choice: 6

----- SALES SUMMARY -----

Total Records : 10
Total Quantity Sold : 500
Total Revenue : Rs. 123456.00
Average Unit Price : Rs. 1367.80
Top Product : Study Table

By Category

Electronics Qty: 145 Revenue: Rs. 87654.00
Furniture Qty: 30 Revenue: Rs. 33990.00
Stationery Qty: 325 Revenue: Rs. 18525.00

Error Handling

To make the application reliable and user-friendly, several validation and exception handling techniques have been implemented.

The system checks for:

- Missing CSV files.
- Permission errors while reading or writing files.
- Invalid or malformed CSV records.
- Incorrect numeric values.
- Invalid menu selections.
- Invalid filter or sort columns.
- File reading and writing errors.
- Missing directories (created automatically).

These validations help prevent runtime errors, improve data integrity, and provide a smooth user experience.

Future Enhancements

The project can be further improved by adding:

- Support for Excel (.xlsx) files.
- JSON data import and export.
- Record editing and deletion features.
- Graphical reports and charts.
- Multi-column sorting and advanced filtering.
- Administrator login system.
- Cloud database integration.
- Web-based version using Flask or Django.
- Graphical User Interface (GUI) using Tkinter.

Conclusion

The **File Handling & Data Processing System** was developed to gain practical experience in Python file handling by solving a real-world data management problem. Through this project, I learned how to read and write CSV files, process structured data efficiently, perform searching, filtering, sorting, and generate analytical reports.

Working on this project also improved my understanding of modular programming, logging, exception handling, and report generation. Features such as automatic sample data creation, sales analytics, exporting reports, and activity logging made the application more practical while enhancing my problem-solving abilities.

Overall, this project strengthened my confidence in building complete Python applications for file handling and data processing, providing valuable hands-on experience that will be useful in future software development projects.